OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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REVIEW

Driving force

Equation of motion $\ddot{x}(t) = -\omega_0^2 x(t) - \gamma\dot{x}(t) + f_0 cos(\omega_d t)$

damping

Intrinsic harmonic Osc

General solution $x(t) = x_c(t) + x_p(t)$

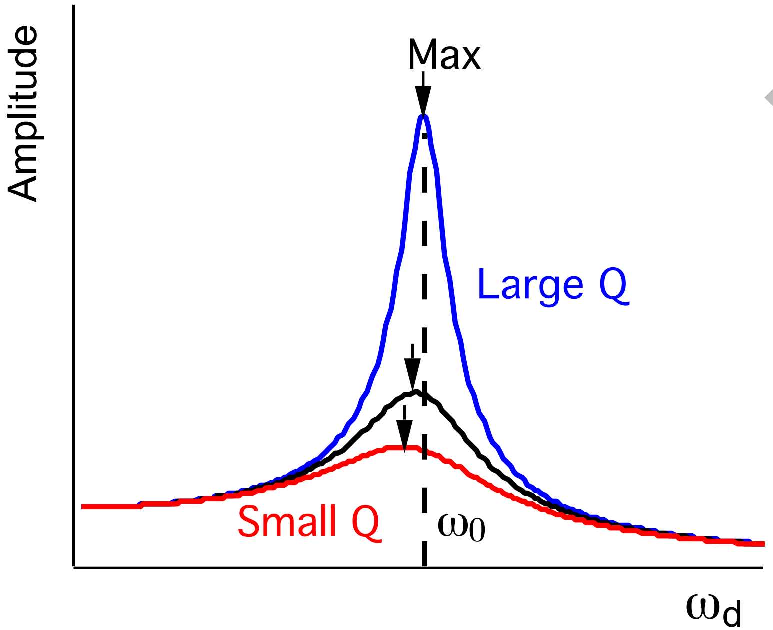
Particular solution $x(t) = Be^{i\omega_d t}$

Characteristic eq. $\left(-\omega_d^2 + i\omega_d\gamma + \omega_0^2\right)Be^{-i\delta} = f_0$

REVIEW

$$\ddot{x}(t) = -\omega_0^2 x(t) - \gamma \dot{x}(t) + f_0 cos(\omega_d t)$$

underdamping $\quad x(t) = Ae^{-\frac{\gamma}{2}t}\, e^{i\omega t} + B_0 e^{i\omega_d (t-\delta/\omega_d)}$



$$B_0 = \frac{f_0}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\omega_d \gamma)^2}}$$

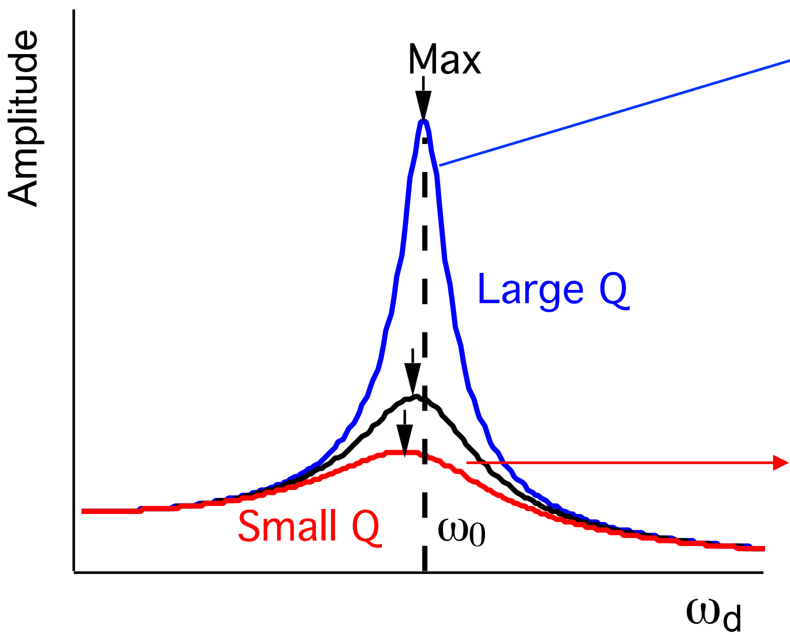
$$Q = \frac{\omega_0}{\gamma} > {}^{1}/_{\sqrt{2}}$$

$$\omega_d \rightarrow \omega_0$$

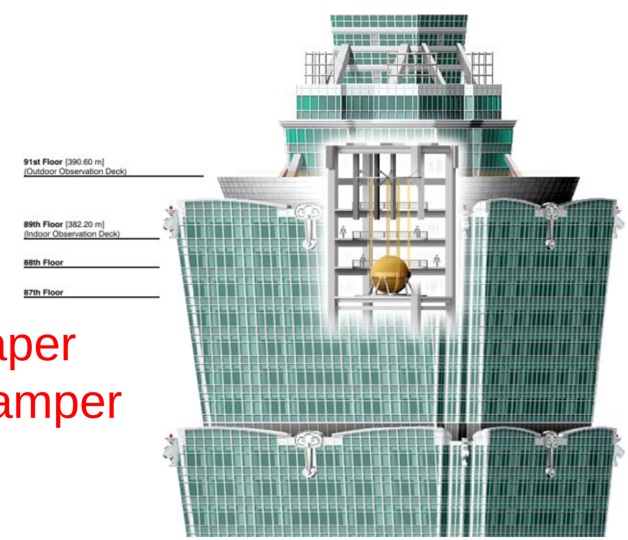
$$B_0 \rightarrow B_{max}$$

REVIEW

$$B_0 = \frac{f_0}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\omega_d \gamma)^2}}$$

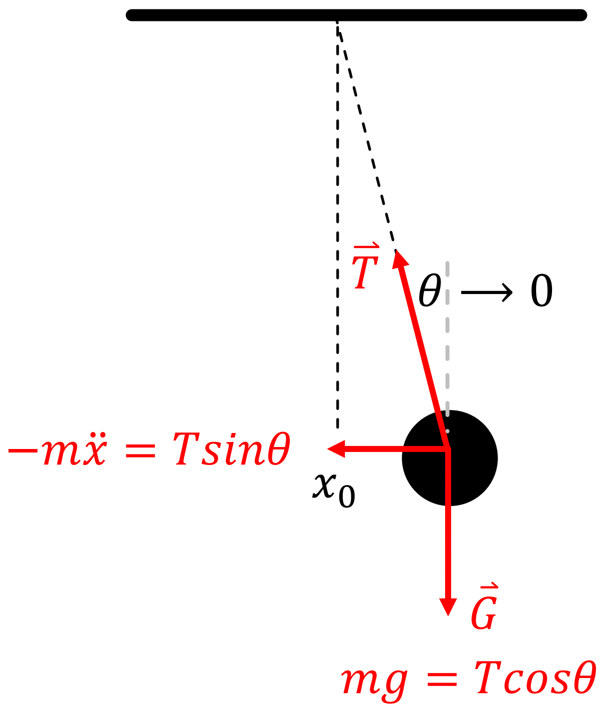


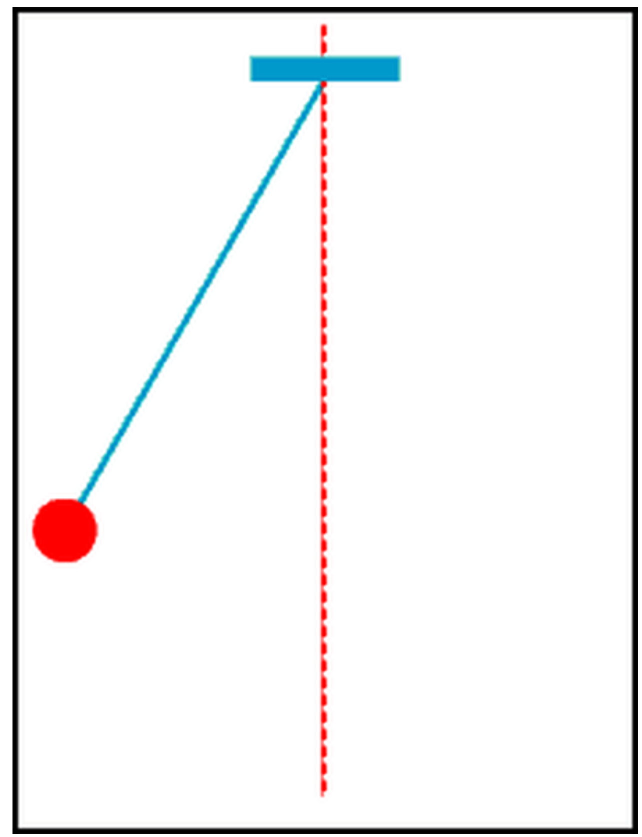
Skyscraper
Wind damper



HOMEWORK

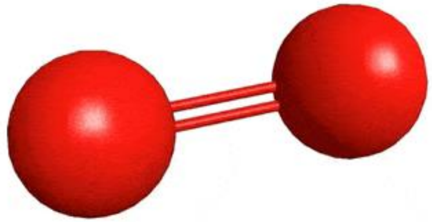
pendulum





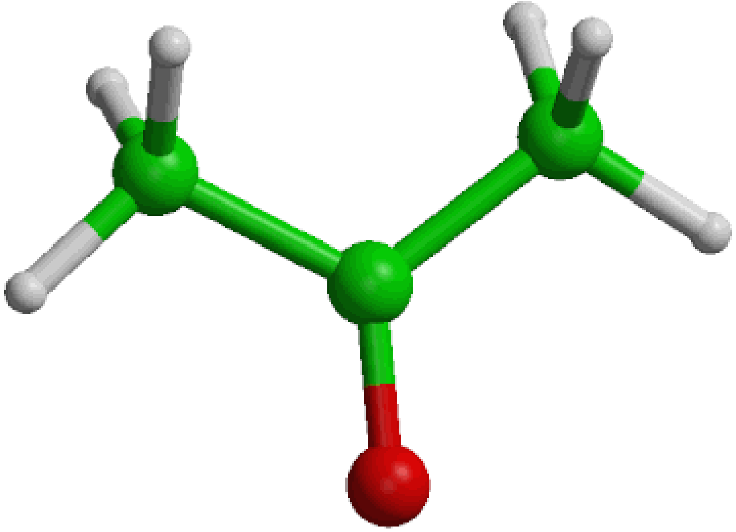
$$-m\ddot{x} = mg\frac{x}{l} \qquad \ddot{x} + \frac{g}{l}x = 0 \qquad \omega_0 = \sqrt{g/l}$$

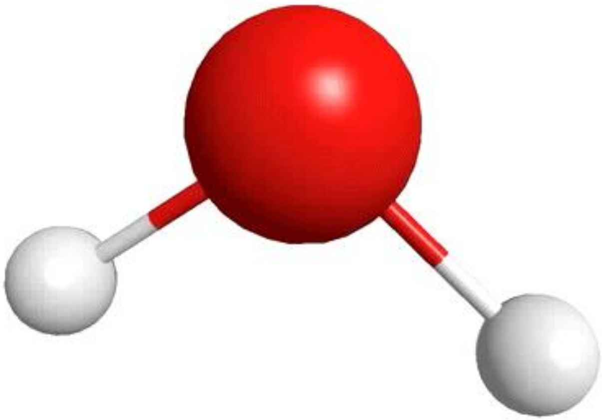
2.1. COUPLED OSCILLATORS



Single oscillator

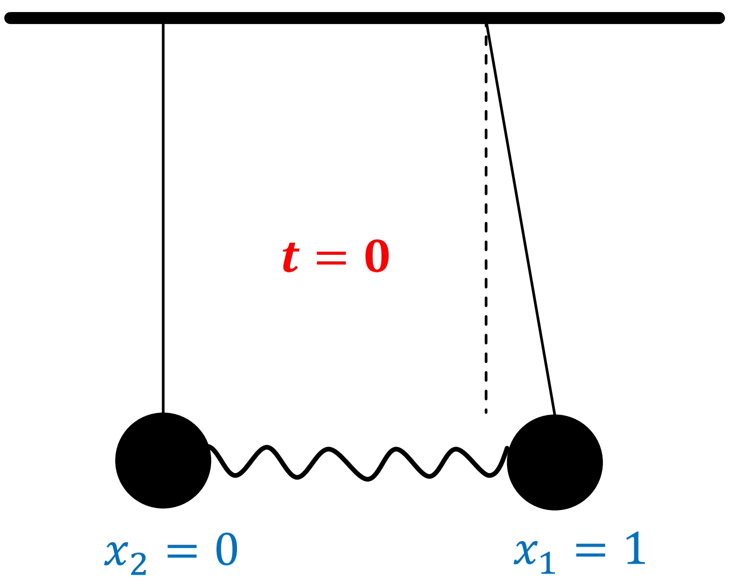
Coupled oscillators



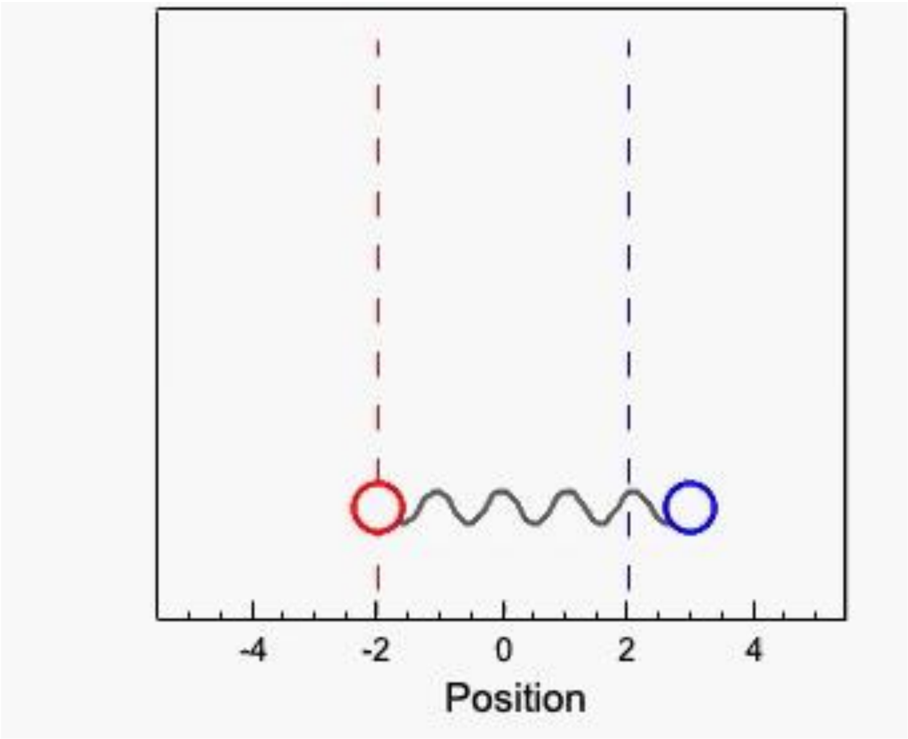


Coupled oscillators

2.1. COUPLED OSCILLATORS

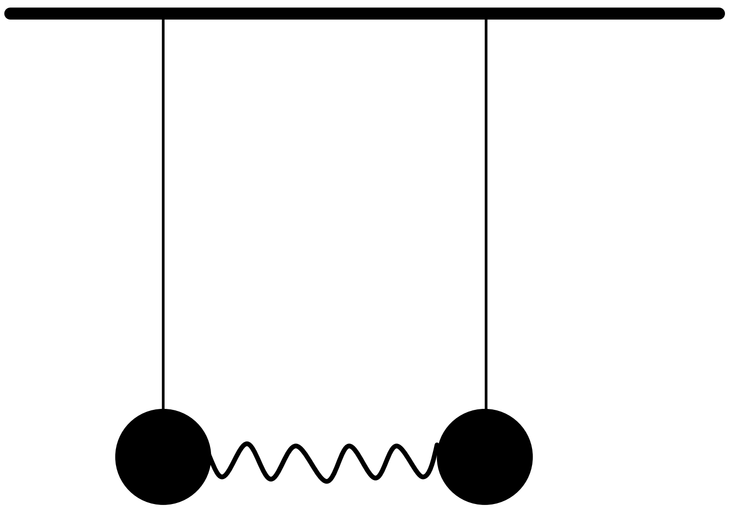


Coupled pendulums



How will these two pendulums move?

2.1. COUPLED OSCILLATORS

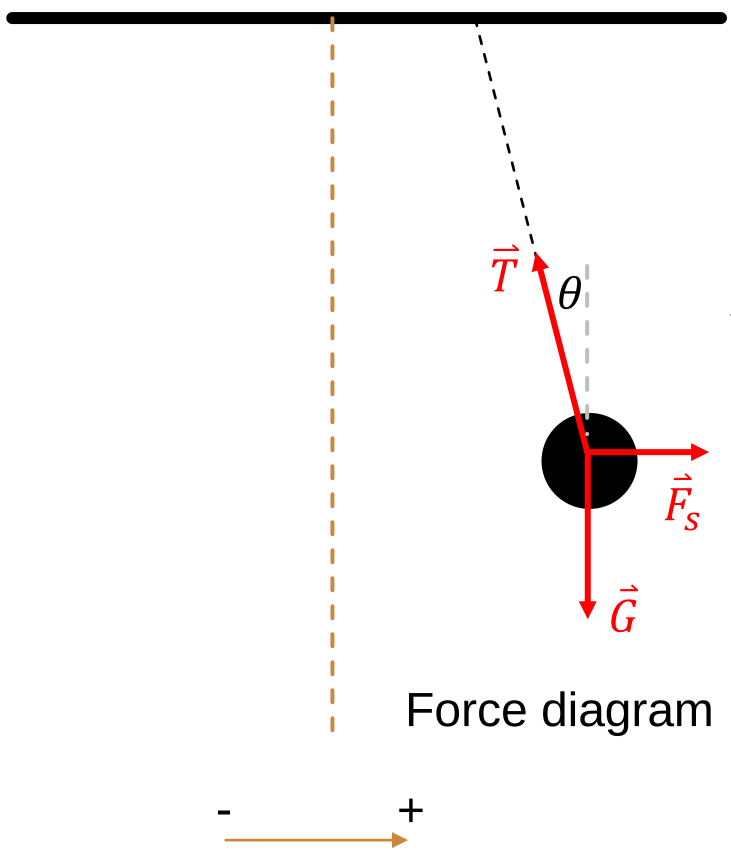


1. Force analysis (Physics)
2. Equation of motion (Math)
3. General solution (Physics)
4. Boundary condition

pendulum $\ddot{x} + \frac{g}{l}x = 0$ $\omega_0 = \sqrt{g/l}$

Coupled pendulums ? ?

2.1. COUPLED OSCILLATORS



Force diagram

Coupled pendulums

$\hat{y}$: $m\ddot{y} = mg - T\cos\theta$

$\hat{x}$: $m\ddot{x} = F_s - T\sin\theta$

Small oscillation approximation: $\theta \longrightarrow 0$

$\cos\theta \sim 1$

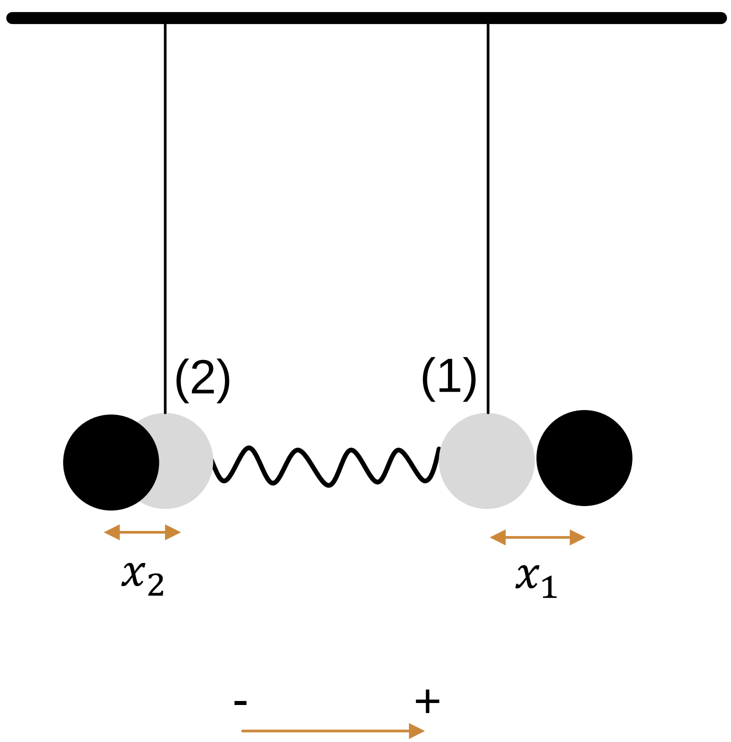
$y_1(t) = y_1(0) \cdot \cos\theta \approx y_1(0)$

$m\ddot{y} \sim 0 \qquad mg = T$

$\sin\theta \sim \theta$

$\hat{x}$: $$m\ddot{x} = F_s - mg\frac{x}{l}$$

2.1. COUPLED OSCILLATORS



Coupled pendulums

Internal coordinates:

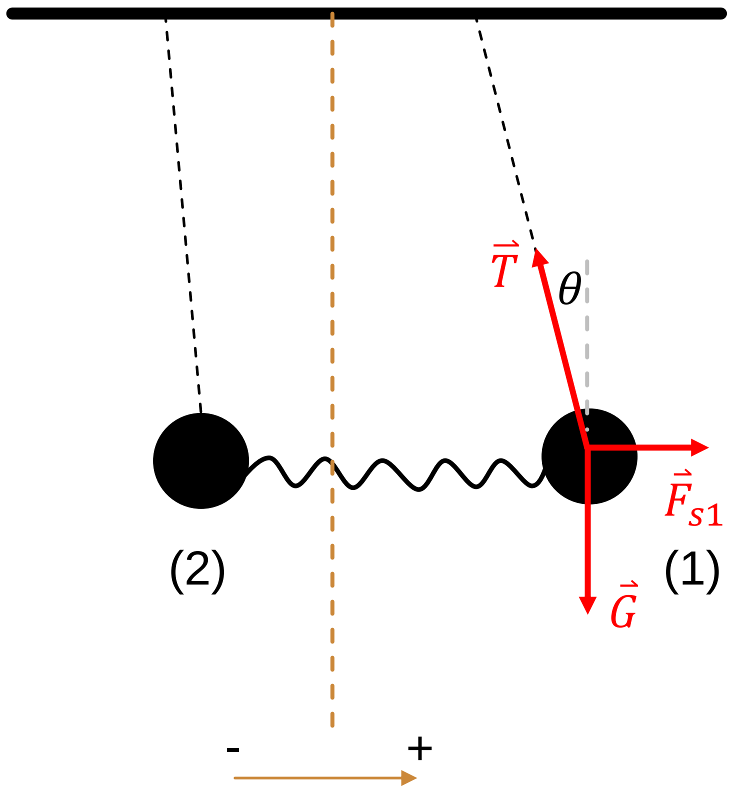
$$F_{s1} = -k(x_1 - x_2)$$

$$F_{s2} = -k(x_2 - x_1)$$

Pay attention to the vectors!

x_1, x_2 are new coordinates from equilibrium positions

2.1. COUPLED OSCILLATORS



Coupled pendulums

$$F_{s1} = -k(x_1 - x_2)$$

$$F_{s2} = -k(x_2 - x_1)$$

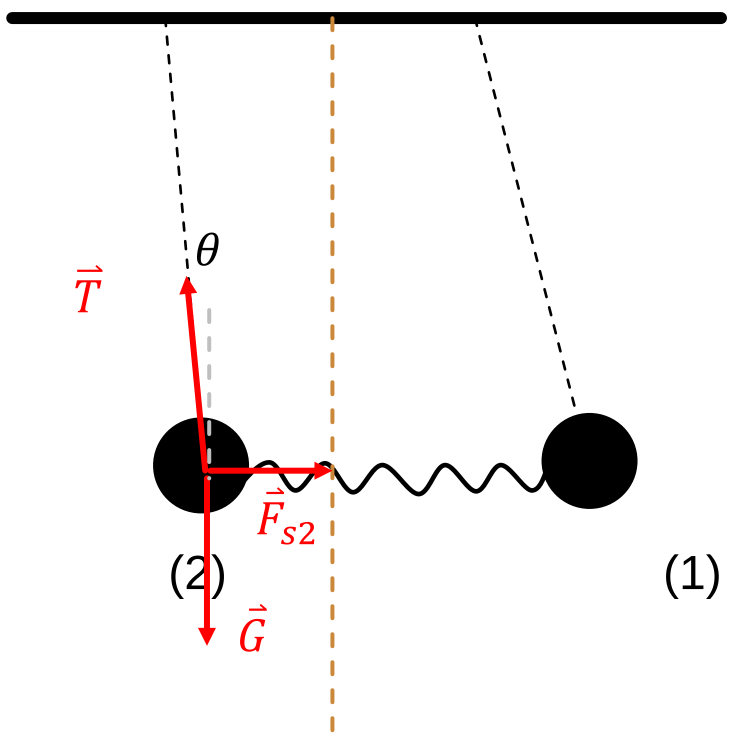
$$\hat{x}: \quad m\ddot{x}_1 = -k(x_1 - x_2) - mg\frac{x_1}{l}$$

Rearrange:

$$m\ddot{x}_1 = -\left(k + \frac{mg}{l}\right)x_1 + kx_2$$

Coupling term

2.1. COUPLED OSCILLATORS



Coupled pendulums

$$F_{s1} = -k(x_1 - x_2)$$

$$F_{s2} = -k(x_2 - x_1)$$

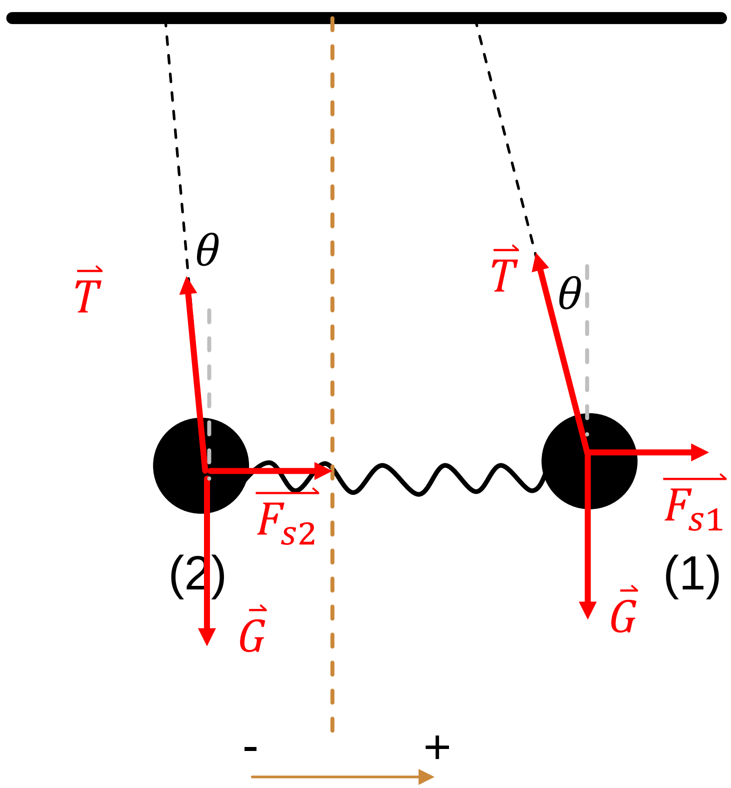
$$\hat{x}: \quad m\ddot{x}_2 = -k(x_2 - x_1) - mg\frac{x_2}{l}$$

Rearrange:

$$m\ddot{x}_2 = -\left(k + \frac{mg}{l}\right)x_2 + kx_1$$

Coupling term

2.1. COUPLED OSCILLATORS



Coupled pendulums

$$m\ddot{x}_1 = -\left(k + \frac{mg}{l}\right)x_1 + kx_2$$

$$m\ddot{x}_2 = kx_1 - \left(k + \frac{mg}{l}\right)x_2$$

Coupling terms

Force diagram

2.1. COUPLED OSCILLATORS

Two equation, two variables

$\hat{x}$:

$$m\ddot{x}_1 = -\left(k + \frac{mg}{l}\right)x_1 + kx_2$$

$$m\ddot{x}_2 = kx_1 - \left(k + \frac{mg}{l}\right)x_2$$

$\hat{x}$:

$$\ddot{x}_1 = -\left(\frac{k}{m} + \frac{g}{l}\right)x_1 + \frac{k}{m}x_2$$

$$\ddot{x}_2 = \frac{k}{m}x_1 - \left(\frac{k}{m} + \frac{g}{l}\right)x_2$$

$$\ddot{\boldsymbol{X}} = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{X}$$

Two equation, two variables

$$\begin{cases} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{cases}$$

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Cramer's rule

$$\begin{array}{ccc} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & \vdots & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n & = & b_n. \end{array}$$

$$\boldsymbol{AX} = \boldsymbol{B} \qquad \boldsymbol{X} = \boldsymbol{A}^{-1}\boldsymbol{B}$$

2.1. COUPLED OSCILLATORS

$$\hat{x}: \quad \ddot{x}_1 = -\left(\frac{k}{m}+\frac{g}{l}\right)x_1 + \frac{k}{m}x_2$$

$$\ddot{x}_2 = \frac{k}{m}x_1 - \left(\frac{k}{m}+\frac{g}{l}\right)x_2$$

$$\ddot{\boldsymbol{X}} = \begin{pmatrix}\ddot{x}_1\\ \ddot{x}_2\end{pmatrix} \qquad \boldsymbol{X} = \begin{pmatrix}x_1\\ x_2\end{pmatrix} \qquad \boldsymbol{M}^{-1}\boldsymbol{K} = \begin{pmatrix}\frac{k}{m}+\frac{g}{l} & -\frac{k}{m}\\ -\frac{k}{m} & \frac{k}{m}+\frac{g}{l}\end{pmatrix}$$

$$\ddot{\boldsymbol{X}} = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{X}$$

2.1. COUPLED OSCILLATORS

$$\ddot{\boldsymbol{X}} = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{X}$$

Trial solution: $\boldsymbol{X}$ $\qquad \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} e^{i(\omega t+\phi)} = \boldsymbol{A}e^{i(\omega t+\phi)}$

$\ddot{\boldsymbol{X}}$ $\qquad -\omega^2 \boldsymbol{A} e^{i(\omega t+\phi)}$

$$\omega^2 \boldsymbol{A} = \boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{A} \qquad (\boldsymbol{M}^{-1}\boldsymbol{K} - \omega^2 \boldsymbol{I})\boldsymbol{A} = \boldsymbol{0}$$

$$\omega^2 \boldsymbol{I} = \begin{pmatrix} \omega^2 & 0 \\ 0 & \omega^2 \end{pmatrix} \qquad \begin{pmatrix} \frac{k}{m} + \frac{g}{l} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Nontrivial solution $\qquad det(\omega^2 \boldsymbol{I} - \boldsymbol{M}^{-1}\boldsymbol{K}) = 0$

2.1. COUPLED OSCILLATORS

Nontrivial solution $$det\left(\boldsymbol{M}^{-1}\boldsymbol{K} - \omega^2\boldsymbol{I}\right) = 0$$

$$\begin{vmatrix} \frac{k}{m} + \frac{g}{l} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega^2 \end{vmatrix} = 0$$

$$\left(\frac{k}{m} + \frac{g}{l} - \omega^2\right)^2 - \left(\frac{k}{m}\right)^2 = 0$$

$$\omega_2^2 = \frac{2k}{m} + \frac{g}{l} \qquad \omega_1^2 = \frac{g}{l}$$

2.1. COUPLED OSCILLATORS

Eq. of Motion $\ddot{\boldsymbol{X}} = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{X}$ Trial solution: $\boldsymbol{X} = \boldsymbol{A}e^{i(\omega t+\phi)}$

$$(\boldsymbol{M}^{-1}\boldsymbol{K} - \omega^2\boldsymbol{I})\boldsymbol{A} = \boldsymbol{0}$$

$$\begin{pmatrix} \frac{k}{m} + \frac{g}{l} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega^2 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \boldsymbol{0}$$

$$\omega_1^2 = \frac{g}{l}$$

$$\frac{k}{m}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \frac{k}{m}\begin{pmatrix} a_1 - a_2 \\ a_2 - a_1 \end{pmatrix} = \boldsymbol{0}$$

$$a_1 = a_2 \qquad \boldsymbol{A} = \alpha\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2.1. COUPLED OSCILLATORS

Eq. of Motion $\ddot{\boldsymbol{X}} = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{X}$ Trial solution: $\boldsymbol{X} = \boldsymbol{A}e^{i(\omega t+\phi)}$

$$(\boldsymbol{M}^{-1}\boldsymbol{K} - \omega^2\boldsymbol{I})\boldsymbol{A} = \boldsymbol{0}$$

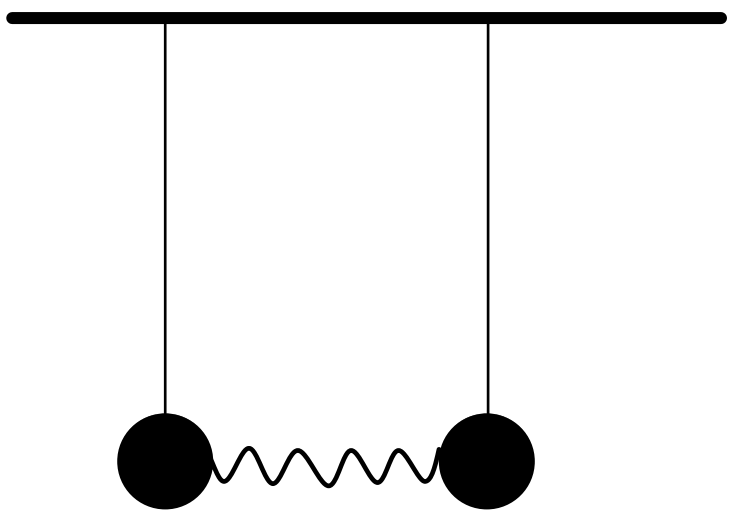
$$\begin{pmatrix} \frac{k}{m} + \frac{g}{l} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \boldsymbol{0}$$

$$\omega_2^2 = \frac{2k}{m} + \frac{g}{l}$$

$$\frac{k}{m}\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \frac{k}{m}\begin{pmatrix} a_1 + a_2 \\ a_2 + a_1 \end{pmatrix} = 0$$

$$a_1 = -a_2 \qquad \boldsymbol{A} = \beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

2.1. COUPLED OSCILLATORS



Coupled pendulums

$$\boldsymbol{X} = \boldsymbol{A}e^{i(\omega_1 t+\phi)} \qquad \omega_1^2 = \frac{g}{l}$$

$$\boldsymbol{A} = A\begin{pmatrix}1\\1\end{pmatrix}$$

$$\boxed{\boldsymbol{X}_a = \begin{pmatrix}x_1\\x_2\end{pmatrix} = \alpha\begin{pmatrix}1\\1\end{pmatrix}e^{i(\sqrt{\frac{g}{l}}t+\phi)}}$$

$$\boldsymbol{X} = \boldsymbol{A}e^{i(\omega_2 t+\phi)} \qquad \omega_2^2 = \frac{2k}{m} + \frac{g}{l}$$

$$\boldsymbol{A} = B\begin{pmatrix}1\\-1\end{pmatrix}$$

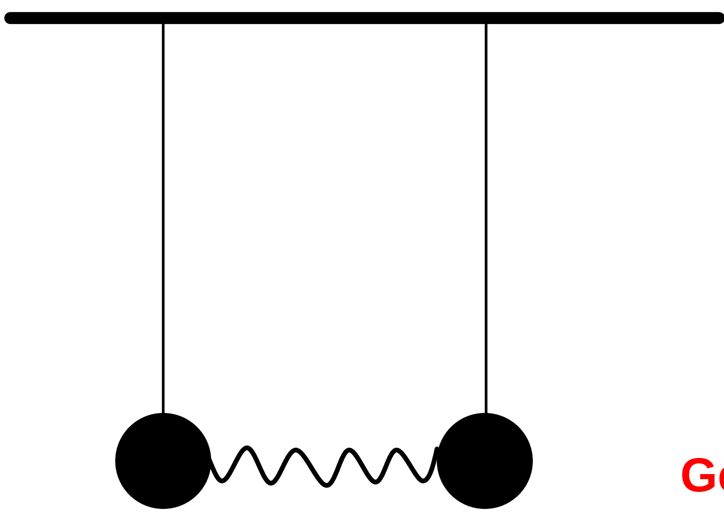
$$\boxed{\boldsymbol{X}_b = \begin{pmatrix}x_1\\x_2\end{pmatrix} = \beta\begin{pmatrix}1\\-1\end{pmatrix}e^{i(\sqrt{\frac{2k}{m}+\frac{g}{l}}t+\phi)}}$$

2.1. COUPLED OSCILLATORS

	Single Harmonic oscillator	Coupled oscillators
Equation of motion	$\ddot{x} + \omega_0^2 x(t) = 0$	$\ddot{\boldsymbol{X}} + \boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{X} = \boldsymbol{0}$
Trial solution	$x(t) = Ae^{i(\omega t+\phi)}$	$\boldsymbol{X} = \boldsymbol{A}e^{i(\omega t+\phi)}$
	$-\omega^2 x(t) + \omega_0^2 x(t) = 0$	$-\omega^2 \boldsymbol{A}e^{i(\omega t+\phi)} + M^{-1}K\boldsymbol{A}e^{i(\omega t+\phi)} = \boldsymbol{0}$
Characteristic equation	$-\omega^2 + \omega_0^2 = 0$	$det(\omega^2 \boldsymbol{I} - M^{-1}K) = 0$
	$\omega^2 = \omega_0^2$	$\omega^2 = \omega_1^2 \quad \omega^2 = \omega_2^2$
General solution	$x(t) = Ae^{i(\omega_0 t+\phi)}$	$\boldsymbol{X} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i(\omega_1 t+\phi)} + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i(\omega_2 t+\phi)}$

2.1. COUPLED OSCILLATORS

Coupled pendulums



$$\boldsymbol{X}_a \begin{cases} x_1(t) = \alpha e^{i\left(\sqrt{\frac{g}{l}}t+\phi\right)} \\ x_2(t) = \alpha e^{i\left(\sqrt{\frac{g}{l}}t+\phi\right)} \end{cases}$$

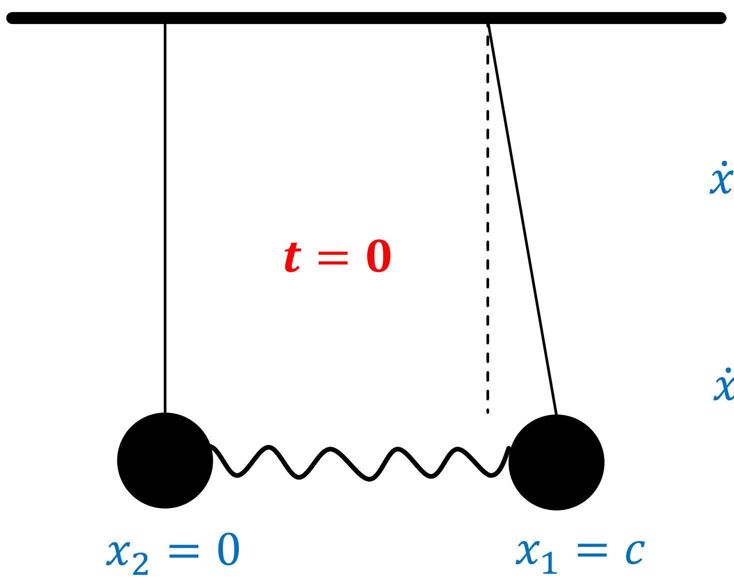
$$\boldsymbol{X}_b \begin{cases} x_1(t) = \beta e^{i\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t+\varphi\right)} \\ x_2(t) = -\beta e^{i\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t+\varphi\right)} \end{cases}$$

General solutions:

$$\boldsymbol{X} \begin{cases} x_1(t) = \alpha\cos\left(\sqrt{\frac{g}{l}}t+\phi\right) + \beta\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t+\varphi\right) \\ x_2(t) = \alpha\cos\left(\sqrt{\frac{g}{l}}t+\phi\right) - \beta\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t+\varphi\right) \end{cases}$$

$\alpha, \beta, \phi, \varphi$ can be determined from initial or boundary conditions

2.1. COUPLED OSCILLATORS



Coupled pendulums

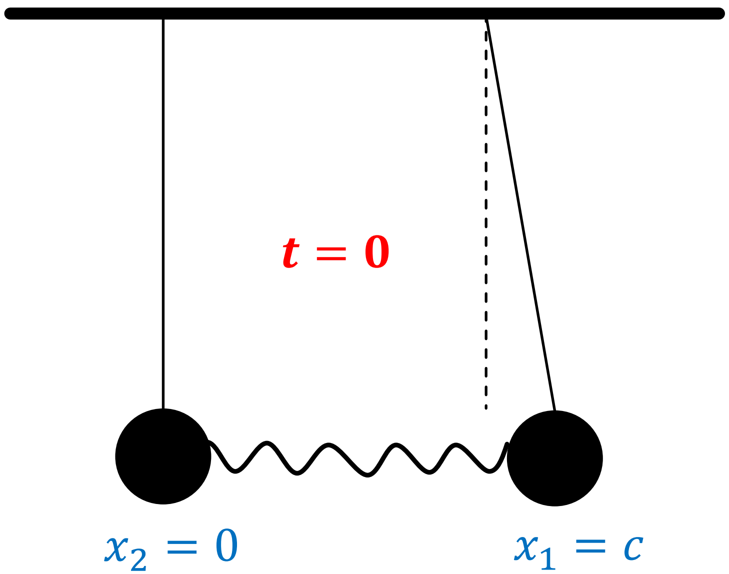
Initial condition:

$$\dot{x}_1(0) = -\alpha\sqrt{\frac{g}{l}}\sin(\phi) - \beta\sqrt{\frac{2k}{m}+\frac{g}{l}}\sin(\varphi) = 0$$

$$\dot{x}_2(0) = -\alpha\sqrt{\frac{g}{l}}\sin(\phi) + \beta\sqrt{\frac{2k}{m}+\frac{g}{l}}\sin(\varphi) = 0$$

$$\Rightarrow \begin{cases} \phi = 0 \\ \varphi = 0 \end{cases}$$

2.1. COUPLED OSCILLATORS



Coupled pendulums

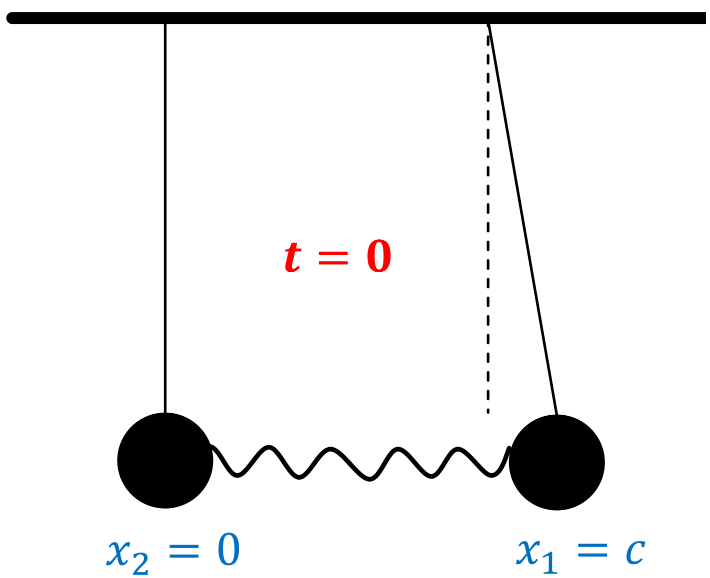
Initial condition:

$$\left.\begin{aligned} x_1(0) &= \alpha + \beta = \mathrm{c} \\ x_2(0) &= \alpha - \beta = 0 \end{aligned}\right\} \alpha = \beta = c/2$$

$$x_1(t) = 0.5c\cos\left(\sqrt{\frac{g}{l}}t\right) + 0.5c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t\right)$$

$$x_2(t) = 0.5c\cos\left(\sqrt{\frac{g}{l}}t\right) - 0.5c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t\right)$$

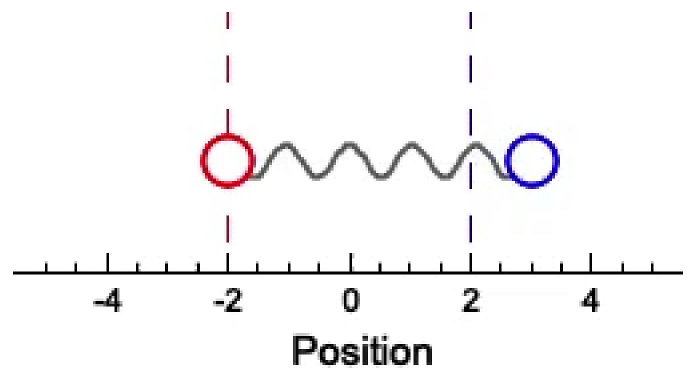
2.1. COUPLED OSCILLATORS



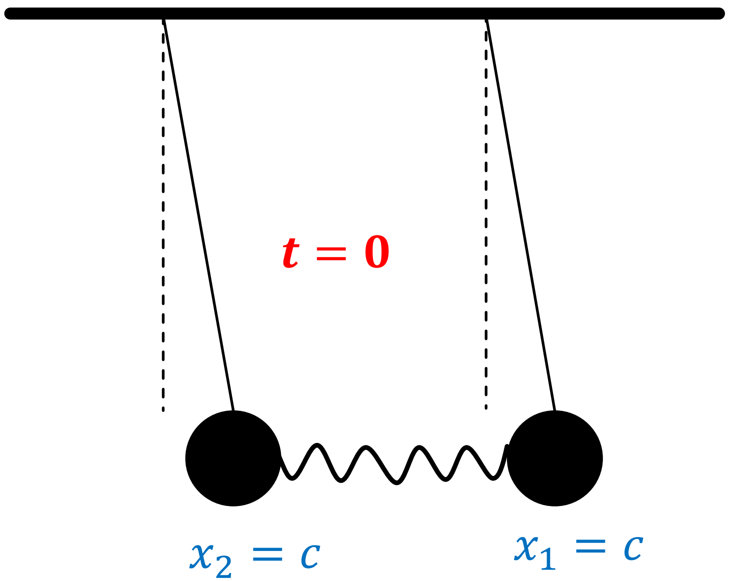
Coupled pendulums

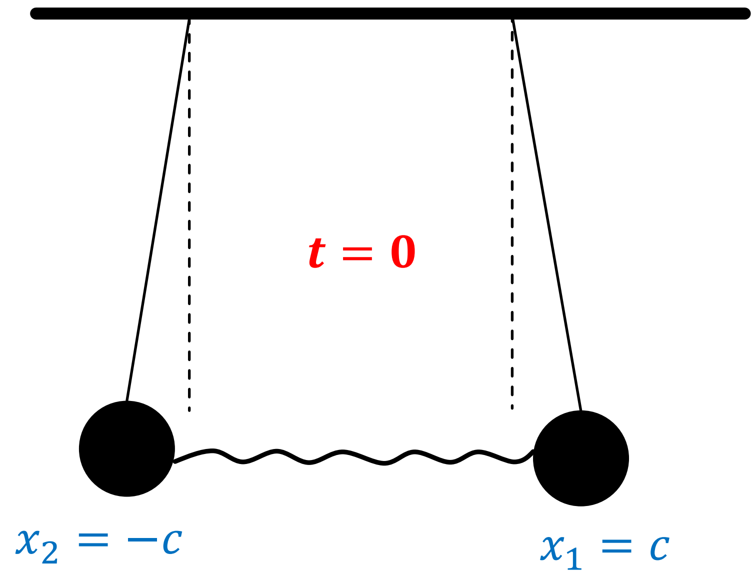
$$x_1(t) = 0.5c\cos\left(\sqrt{\frac{g}{l}}t\right) + 0.5c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t\right)$$

$$x_2(t) = 0.5c\cos\left(\sqrt{\frac{g}{l}}t\right) - 0.5c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t\right)$$



2.2. NORMAL MODES

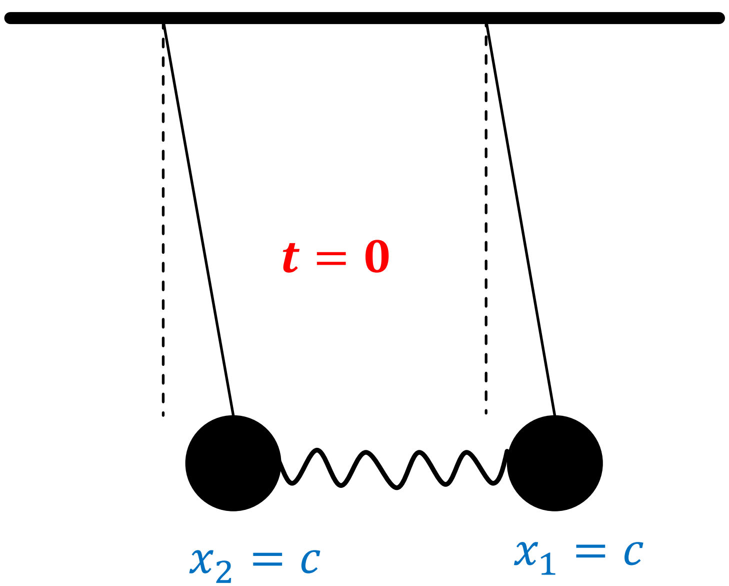




Some special initial conditions ???

2.2. NORMAL MODES

$$\dot{x}_1(0) = \dot{x}_2(0) = 0$$



Coupled pendulums

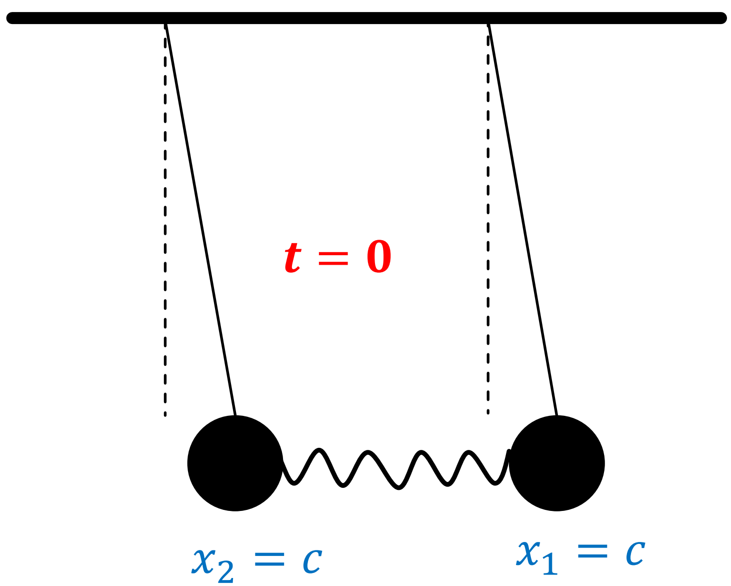
$$\begin{cases} x_1(t) = \alpha \cos\left(\sqrt{\frac{g}{l}}\,t\right) + \beta \cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}}\,t\right) \\ x_2(t) = \alpha \cos\left(\sqrt{\frac{g}{l}}\,t\right) - \beta \cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}}\,t\right) \end{cases}$$

Initial conditions A:

$$\left.\begin{array}{l} x_1(0) = x_2(0) = \mathrm{c} \\ x_1(0) = \alpha + \beta \\ x_2(0) = \alpha - \beta \end{array}\right\} \quad \alpha = c;\ \beta = 0$$

$$\begin{cases} x_1(t) = c\cos\left(\sqrt{\frac{g}{l}}\,t\right) \\ x_2(t) = c\cos\left(\sqrt{\frac{g}{l}}\,t\right) \end{cases}$$

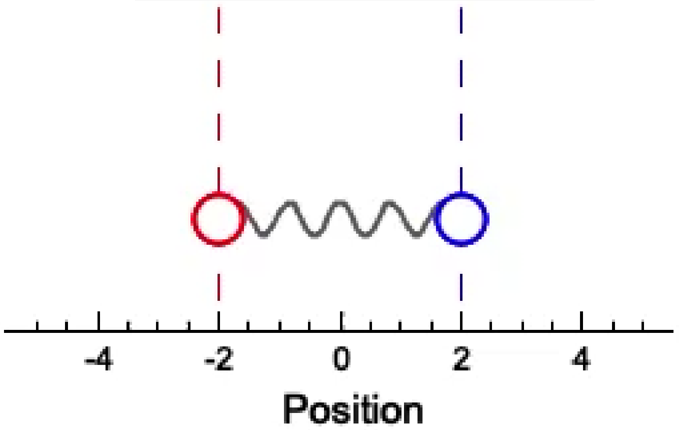
2.2. NORMAL MODES



Coupled pendulums

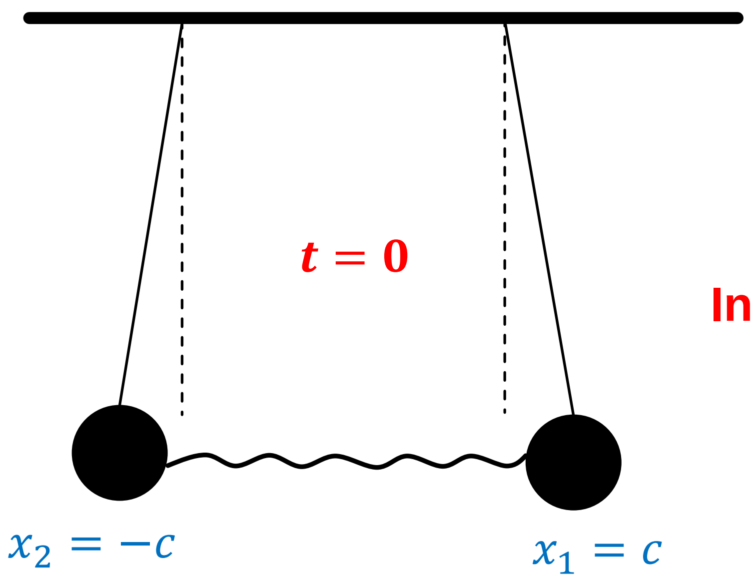
$$x_1(t) = c\cos(\sqrt{\frac{g}{l}}t)$$

$$x_2(t) = c\cos(\sqrt{\frac{g}{l}}t)$$



2.2. NORMAL MODES

$$\dot{x}_1(0) = \dot{x}_2(0) = 0$$



Coupled pendulums

$$\begin{cases} x_1(t) = \alpha\cos\left(\sqrt{\frac{g}{l}}t\right) + \beta\cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}}t\right) \\ x_2(t) = \alpha\cos\left(\sqrt{\frac{g}{l}}t\right) - \beta\cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}}t\right) \end{cases}$$

Initial conditions B:

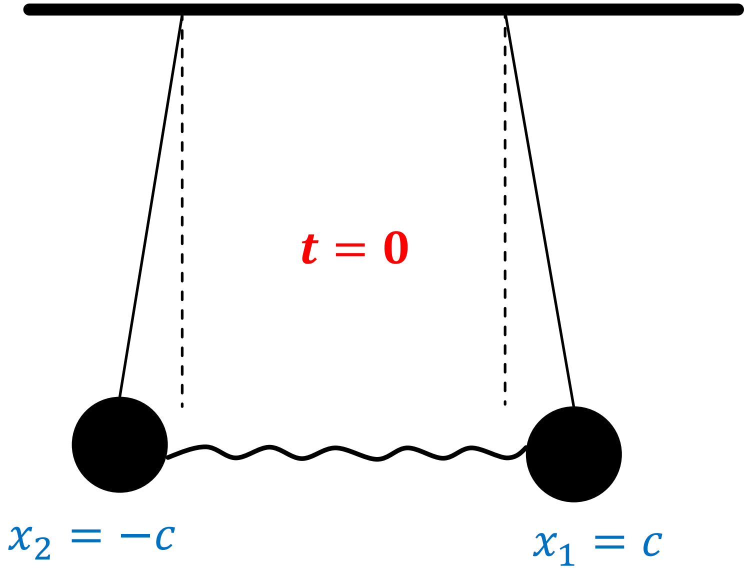
$$\left.\begin{array}{l} x_1(0) = -x_2(0) = \mathrm{c} \\ x_1(0) = \alpha + \beta \\ x_2(0) = \alpha - \beta \end{array}\right\} \alpha = 0;\ \beta = c$$

$$\begin{cases} x_1(t) = c\cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}}t\right) \\ x_2(t) = -c\cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}}t\right) \end{cases}$$

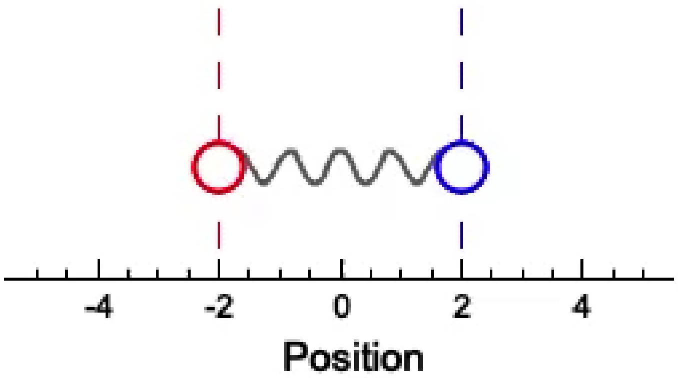
2.2. NORMAL MODES

$$x_1(t) = c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}\,t\right)$$

$$x_2(t) = -c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}\,t\right)$$



Coupled pendulums



2.2. NORMAL MODES

Normal Mode: A normal mode of an oscillating system is a pattern of motion in which all parts of the system move sinusoidally with the same frequency and with a fixed phase relation.

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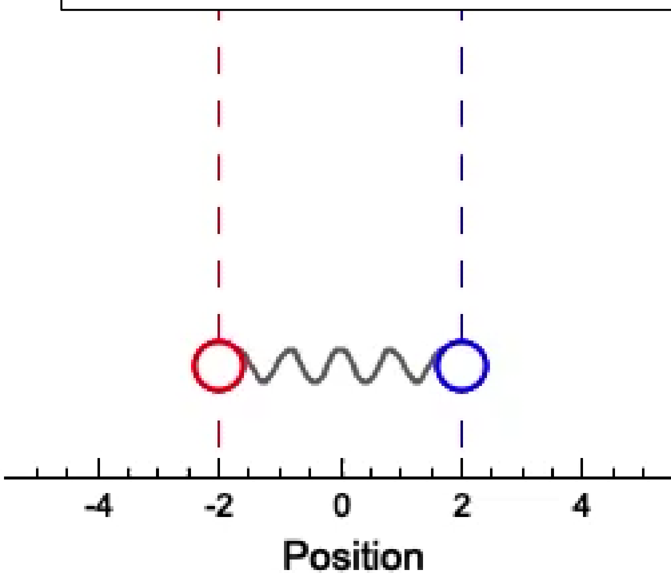
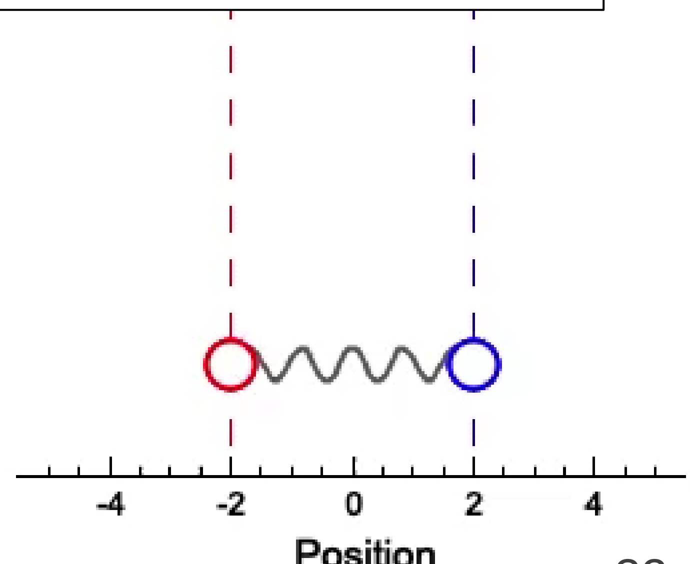
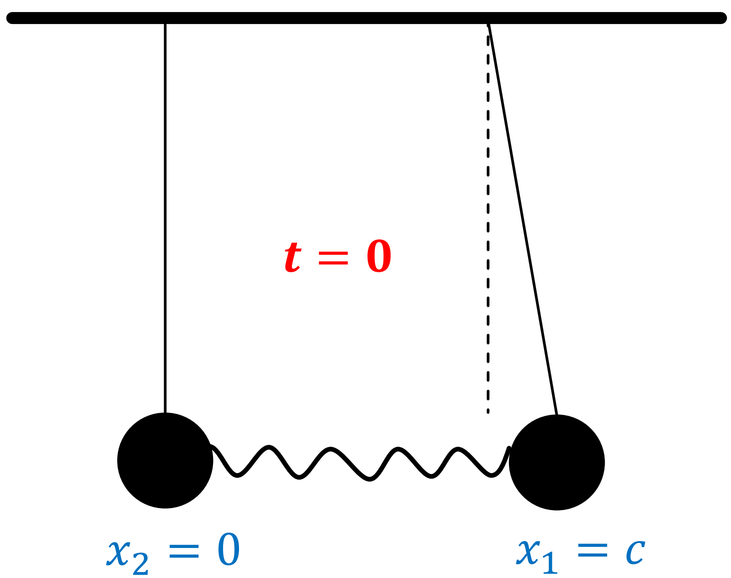
The most general motion of a system is a superposition of its normal modes.





2.2. NORMAL MODES



Coupled pendulums

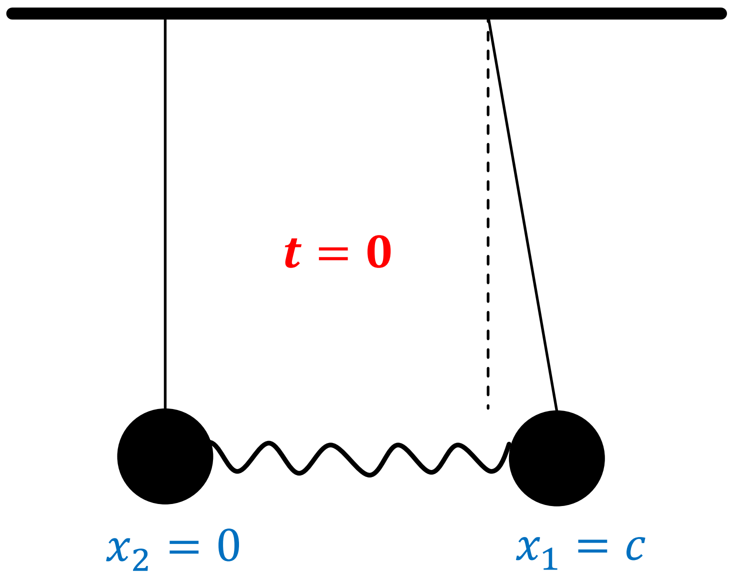
Random initial condition:

$$x_1(t) = 0.5c\cos\left(\sqrt{\frac{g}{l}}t\right) + 0.5c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t\right)$$

$$x_2(t) = 0.5c\cos\left(\sqrt{\frac{g}{l}}t\right) - 0.5c\cos\left(\sqrt{\frac{2k}{m}+\frac{g}{l}}t\right)$$

The most general motion of a system is a superposition of its normal modes.

2.2. NORMAL MODES-BEATING



Coupled pendulums

$$\omega_1^2 = \frac{g}{l} \qquad \omega_2^2 = \frac{2k}{m} + \frac{g}{l}$$

$$x_1(t) = 0.5c[\cos(\omega_1 t) + \cos(\omega_2 t)]$$

$$x_2(t) = 0.5c[\cos(\omega_1 t) - \cos(\omega_2 t)]$$

$$x_1(t) = \mathrm{c} \cdot \cos\left(\frac{\omega_1+\omega_2}{2} t\right) \cos\left(\frac{\omega_1-\omega_2}{2} t\right)$$

$$x_2(t) = -c \cdot \sin\left(\frac{\omega_1+\omega_2}{2} t\right) \sin\left(\frac{\omega_1-\omega_2}{2} t\right)$$

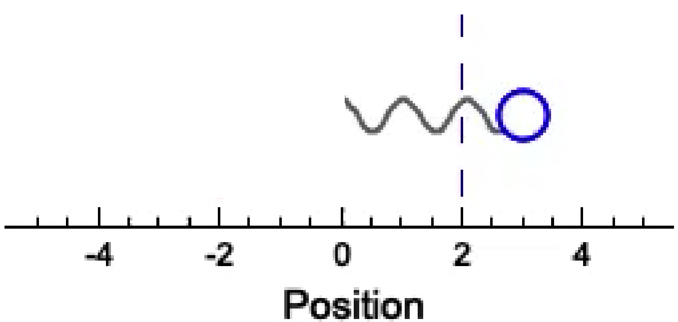
2.2. NORMAL MODES-BEATING

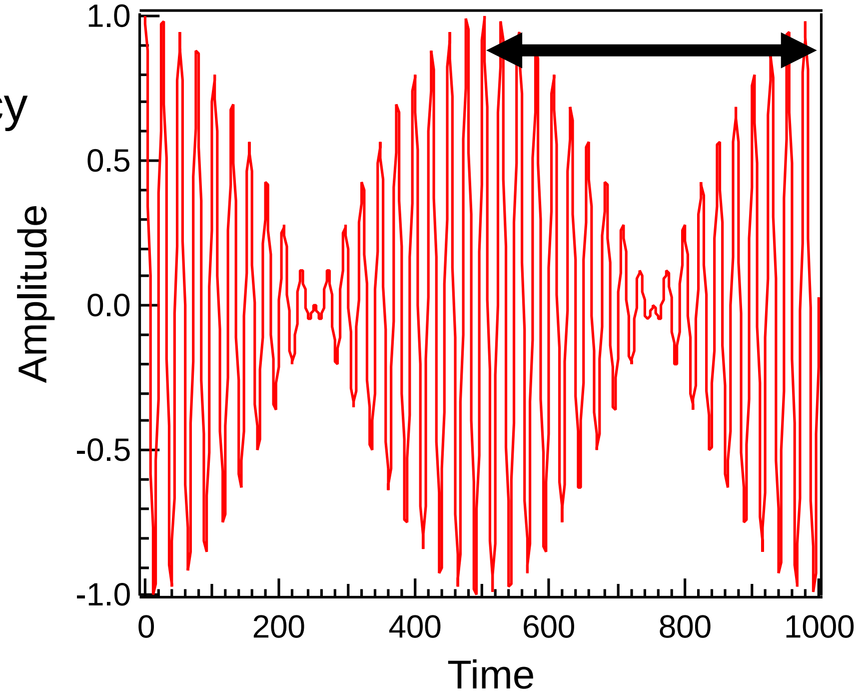
$$x_1(t) = \mathrm{c} \cdot \cos\left(\frac{\omega_1+\omega_2}{2}t\right) \cos\left(\frac{\omega_1-\omega_2}{2}t\right)$$

Beat frequency

If $\omega_1 = 0.95\omega_2$

$T = 2\pi/(\omega_1 - \omega_2)$





Beating phenomenon: interference pattern between two different frequencies
Envelope function

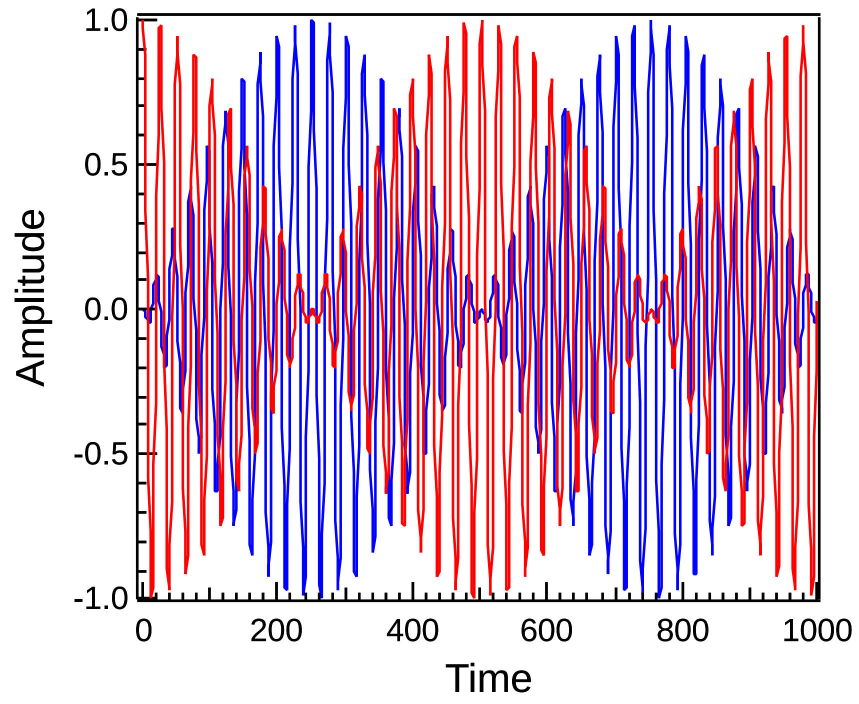
2.2. NORMAL MODES-BEATING

$$x_1(t) = \mathrm{c} \cdot \cos\left(\frac{\omega_1+\omega_2}{2}t\right)\cos\left(\frac{\omega_1-\omega_2}{2}t\right)$$

$$x_2(t) = -c \cdot \sin\left(\frac{\omega_1+\omega_2}{2}t\right)\sin\left(\frac{\omega_1-\omega_2}{2}t\right)$$

If $\omega_1 = 0.95\omega_2$

$x_2(t)$

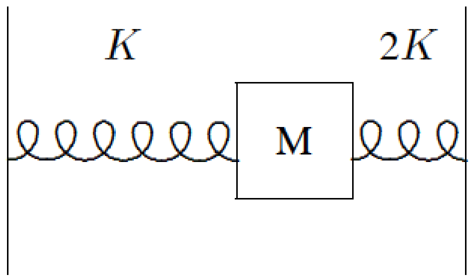


Energy flow
Amplitude varying
Not harmonic

Application: detecting high frequency vibration

HOMEWORK

A block of mass M slides without friction between two springs of spring constant K and $2K$, as shown. The block is constrained to move only left and right on the paper, so the system has only one degree of freedom.



Calculate the oscillation angular frequency. If the velocity of the block when it is at its equilibrium position is v, calculate the amplitude of the oscillation.

HOMEWORK

$$\frac{d^2}{dt^2}\,z(t) + \Gamma\,\frac{d}{dt}\,z(t) + \omega_0^2\,z(t) = \mathcal{F}(t)/m\,, \qquad (2.16)$$

where

$$\mathcal{F}(t) = F_0 e^{-i\omega_d t}\,.$$

Prove, ~~just using linearity, without using the explicit solution~~, that the steady state solution to (2.16) must be proportional to F_0.

$$\ddot{z} + \Gamma\dot{z} + \omega_0^2 z - \frac{F_0}{m} e^{-i\omega_d t} = 0$$